ACTIVITY 6.
Identification Of Electrical Machines, Its Parts And Testing.

Electromechanical Machines:

Electromechanical machines are devices that convert electrical energy to mechanical energy.

Examples include: electric motors, generators, transformers, and relays.

1. **Electric motors:** these devices convert electrical energy into mechanical energy. They work on the principle of electromagnetic induction.
2. **Generators:** Generators are devices that convert mechanical energy into electrical energy. They work on the principle of electromagnetic induction, where a magnetic field is rotated within a coil of wire, inducing an electrical current in the wire.
3. **Transformers:** transformers are devices that transfer electrical energy from one circuit to another through electromagnetic induction. They are used to increase or decrease the voltage of electrical power.
4. **Relays:** relays are devices that use an electrical signal to switch a mechanical device on or off. They work by using electromagnetic field to control the opening and closing of electrical contacts, allowing them to control the flow of electrical power.

Direct current motors

Working principle: DC motors work on the principle of electromagnetism,where a magnetic field is produced by the interaction of an electric current with a magnetic field.

The interaction between the magnetic fields creates a force that rotates the motor shaft.

COMPONENTS:
The stator (stationary part)
The rotor (rotating part)
[for brushed dc motors]
[Commutators]
[Brushes]
Bearings
Armature
Field windings

The stator: the stationary part of the motor that houses the field windings. It creates a magnetic field that interacts with the rotor to produce torque.
The rotor: the rotating part contains the armature windings. When the stator magnetic fields interacts with the rotor magnetic field, torque is produced, causing the rotor to rotate.
Commutator: it reverses the direction of the current in the armature winding as the rotor rotates. This allows the magnetic field produced by the armature to interact with the magnetic field produce continuous rotation.
Brushes: conductive elements that make contact with the commutator and supply electrical current to the armature windings.
Brushes wear out over time and need to be replaced periodically.
Armature: The armature is a cylindrical core that houses the armature windings .
The armature windings are made up of copper wire and produce the magnetic field that interacts with the stator magnetic field to produce torque .
Field windings: The field windings are coils of wire that create the magnetic field in the stator .
The strength of the magnetic field can be adjusted bu varying the amount of electrical current flowing through the field windings .

Bearings: Bearings are used to support the rotor and allows it to rotate smoothly .
Made of metal and designed to withstand the high spreads and loads of the motor .

DC motors can be classified into several types based on their construction and operation:

1. Brushed DC motors: use commutators and brushes
2. Brushless DC motors: use electronic commutation to control the direction of the current in the rotor windings.
3. Series DC motors: they have the stator and stator and rotor windings connected in series. It allows for high starting torque but low speed control.
4. Shunt DC motors: Shunt DC motors have the stator and rotor windings connected in parallel. Allows for better speed control. But lower starting torque.
5. Compound DC motors: DC motors that combine the characteristics of series and shunt motors, with both series and shunt windings connected to the same armature.
6. Permanent magnet DC motors: use permanent magnet in DC motors to produce the magnetic field, eliminating the need for a separate field winding.

Functions of armature core:

1. Provides a path for the magnetic field.
2. Supports the armature windings.
3. Minimizes energy losses.
4. Improves heat dissipation.

AC MOTORS:

In AC motors, AC is supplied to the stator windings of the motor in some AC machines, currents induced in the rotor windings by electromagnetic induction causing a magnetic field. The two magnetic fields interacts and the rotor rotates.

Types of AC motors/generators
1. 3-phase motor/generator
2. Single phase motor/generator

Universal motors: motors that take both AC and DC.
Commutation refers to changing from AC to DC..

Transformers:
Transformers are electrical devices that are used to transfer electric current electrical energy from one circuit to another through electromagnetic induction.
They work using the principle of Faraday's law of electromagnetic induction, which states that a changing magnetic field can induce a voltage in a coil of wire.

Transformers are used to step-up or step-down the voltage of the AC power supply, which is necessary to transmit power over long distances and to reduce the voltage to a safe level for use in electronic devices.